

A PARABOLIC-RENORMALIZATION CONSTANT GOVERNING THE SHAPE OF THE $1/q$ BULBS OF THE MANDELBROT SET

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ABSTRACT. To first order, the satellite component of the Mandelbrot set attached to the main cardioid at internal angle p/q is a round disc of diameter proportional to q^{-2} , the Ford-circle law. We study the first correction. Writing the multiplier of the bifurcating q -cycle as a power series in the rescaled parameter $u = q^2\varepsilon$, $\lambda = e^{2\pi ip/q}e^\varepsilon$, we prove that its quadratic coefficient is $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$, where $\iota_{p/q}$ is the holomorphic index of the parabolic fixed point of f^q , $f(z) = e^{2\pi ip/q}z + z^2$, and that $\kappa(p/q) + \frac{1}{2}$ is itself the index of the q -fold rotation quotient of f^q . An exact normal-form recursion evaluates κ in $O(q^2)$ operations without loss of precision. For $p = 1$ we find

$$\kappa_0 := \lim_{q \rightarrow \infty} \kappa(1/q) = 0.023825887402200569000 \dots - 0.052304659114003666317 \dots i,$$

with $\kappa(1/q) = \kappa_0 - \frac{i}{2\pi q} + O(q^{-2})$, and we identify κ_0 , to 27 digits, with $\iota(\mathcal{P}_0) - \frac{1}{2}$, where $\mathcal{P}_0$ is the parabolic renormalization (upper horn map) of $z + z^2$. From the horn map we compute κ_0 to 32 digits. The same horn map, multiplied by the Inou–Shishikura phase $e^{-2\pi i/\alpha}$, predicts the bounded- p limits of κ and of the bulb size itself; in degree three the horn map of the cubic cusp germ plays the same role. The term $-\frac{i}{2\pi q}$, indeed the whole $O(1/q)$ part of the multiplier function, comes from the exact phase, a Möbius function of u . Iterating the construction along continued fractions with large partial quotients gives a hierarchy of constants that contracts, by a factor 0.042 per level, to $\kappa_* = 0.011929183413 - 0.037455493752 i$, attached to the fixed point of parabolic renormalization composed with complex conjugation; its first three levels are confirmed on bulbs. No integer relation with standard constants is found.

1. INTRODUCTION

Let $\lambda \in \mathbb{D}$ parametrize the main cardioid of the Mandelbrot set M by the multiplier of the attracting fixed point, $c(\lambda) = \lambda/2 - \lambda^2/4$; equivalently we work with the family

$$f_\lambda(z) = \lambda z + z^2,$$

which is affinely conjugate to $z^2 + c(\lambda)$. At each root of unity $\lambda_0 = e^{2\pi ip/q}$, $\gcd(p, q) = 1$, a satellite hyperbolic component $W_{p/q}$ of period q is attached to the cardioid [2]. Its size is of order q^{-2} [7]; more precisely, to first order in $1/q$ the component is a round disc whose diameter is $2|c'(\lambda_0)|q^{-2} = 2\sin(\pi p/q)q^{-2}$ in the parameter plane, an estimate derived analytically by Fowler and McGuinness [5]. The diameters q^{-2} and the tangency pattern of the $W_{p/q}$ are those of the Ford circles [6], and this is the leading-order content of the familiar analogy between the Mandelbrot set and the Farey tessellation.

This paper is about the *first correction* to that law, which to our knowledge has not been studied before, and about one constant that it produces. Put $\lambda = \lambda_0 e^\varepsilon$ and let $\rho_{p/q}(\varepsilon)$ be the

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multiplier of the q -cycle of f_λ that bifurcates from the fixed point 0. In the bulb-scale variable $u = q^2 \varepsilon$,

$$(1) \quad R_{p/q}(u) := \rho_{p/q}(u/q^2) = \sum_{k \geq 0} r_k(p/q) u^k.$$

The component $W_{p/q}$ is, up to the conformal change $u \mapsto c(\lambda_0 e^{u/q^2})$, the set $\{|R_{p/q}(u)| < 1\}$. We show that $r_0 = 1$ and $r_1 = -1$ for every p/q ; hence to first order $W_{p/q}$ is the disc $|u - 1| < 1$, which is the Ford law. The first coefficient that sees the arithmetic of p/q is

$$\kappa(p/q) := r_2(p/q).$$

Results.

- (1) *Index formula* (Theorem 2.1). $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$, where $\iota_{p/q} = \text{Res}_{z=0} dz/(z - f_{\lambda_0}^q(z))$ is the holomorphic index of the parabolic fixed point. In particular $\kappa(p/q)$ lies in the cyclotomic field $\mathbb{Q}(\zeta_q)$ and $p \mapsto \kappa(p/q)$ is a Galois orbit.
- (2) *Normal form* (Theorem 3.1 and Corollary 3.2). If $N(z) = \lambda_0 z(1 + b_1 z^q + b_2 z^{2q})$ is the resonant normal form of f_{λ_0} , then $q \iota_{p/q} = (q^2 - 1)/2 + b_2/b_1^2$. Moreover $\kappa(p/q) = \iota(\Phi_{p/q}) - \frac{1}{2}$, where $\Phi_{p/q}(W) = N^q(W^{1/q})^q$ is the rotation quotient of f^q . The recursion for b_1, b_2 gives κ exactly in $O(q^2)$ operations and is free of cancellation.
- (3) *The constant* (Numerical result 4.1). The limit $\kappa_0 = \lim_{q \rightarrow \infty} \kappa(1/q)$ exists numerically, with $\kappa(1/q) = \kappa_0 - \frac{i}{2\pi q} + O(q^{-2})$; the bulbs determine κ_0 to 27 digits.
- (4) *Identification* (Theorem 5.2 and Numerical result 5.3). Let $E(Z) = Z + \sum_{n \geq 1} a_n e^{2\pi i n Z}$ be the upper horn map of $z + z^2$ and $\mathcal{P}_0(W) = W \exp(2\pi i \sum a_n W^n)$ the corresponding parabolic germ. Then $\iota(\mathcal{P}_0) - \frac{1}{2} = a_2/(2\pi i a_1^2)$, and this number equals κ_0 to 27 digits. From the horn map,

$$\begin{aligned} \text{Re } \kappa_0 &= 0.023825887402200569000729189645402 \dots, \\ \text{Im } \kappa_0 &= -0.052304659114003666316522486207571 \dots \end{aligned}$$

- (5) *Renormalization* (Section 6). The identification is one instance of a broader picture in which the rotation quotient $\Phi_{p/q}$ is a near-parabolic renormalization of $f_{p/q}$. This picture predicts, with no free parameters, the limits of $\kappa(p/q)$ and of the bulb size along every sequence of bounded p , and the analogous constants for $z^3 + c$. The predicted values of κ agree with the bulb data to 10^{-8} in degree two and $5 \cdot 10^{-6}$ in degree three, and the predicted bulb sizes to $9 \cdot 10^{-7}$. The exact phase is a Möbius function of u , and it accounts for the universal term $-\frac{i}{2\pi pq}$ and for the whole $O(1/q)$ part of $R_{p/q}$ (Section 6.4). Iterating the construction along continued fractions with large partial quotients gives a hierarchy $\kappa_0 = \kappa_1, \kappa_2, \dots$ that contracts by a factor 0.042 per level towards a second constant $\kappa_* = 0.011929183413 - 0.037455493752i$; the second and third levels are confirmed on bulbs, to $3 \cdot 10^{-10}$ and $1.2 \cdot 10^{-6}$.

Items (1) and (2) are theorems. Items (3)–(5) are numerical results, obtained with rigorous or cross-validated arithmetic and stated with their error estimates. We do not prove the identity $\kappa_0 = \iota(\mathcal{P}_0) - \frac{1}{2}$; Section 6 explains why we expect a proof along the lines of the Inou–Shishikura theory of near-parabolic renormalization [8].

Why this constant matters. The number κ_0 controls the shape of the $1/q$ -bulbs, which accumulate at the cusp $c = \frac{1}{4}$ of M ; for instance the diameter proxy of Definition 2.4 satisfies

$$d(1/q) = 2\pi G_0 q^{-3}(1 + O(q^{-1})), \quad G_0 = 1.0264153852 \dots,$$

and G_0 is determined by the same horn map (Table 3). More broadly, $\kappa(p/q)$ is the first correction to the Ford-circle description of M . Numerically, its dependence on p/q is governed, after a transient, by the arithmetic of the modular inverse $p^{-1} \bmod q$, and it is singular at every rational [15]. The renormalization picture explains that dependence, and κ_0 is its simplest instance.

2. SETTING AND THE INDEX FORMULA

Fix p/q in lowest terms, $\lambda_0 = e^{2\pi i p/q}$, and write $f = f_{\lambda_0}$. Since f has a single critical point, the parabolic fixed point 0 has exactly one cycle of q attracting petals, so

$$(2) \quad f^q(z) = z + a z^{q+1} + O(z^{q+2}), \quad a = a(p/q) \neq 0.$$

The holomorphic index [12, §12] of 0 as a fixed point of f^q is

$$\iota_{p/q} := \frac{1}{2\pi i} \oint \frac{dz}{z - f^q(z)} = \operatorname{Res}_{z=0} \frac{dz}{z - f^q(z)},$$

and its résidu itératif [3, 1] is $\text{résit} = \frac{q+1}{2} - \iota_{p/q}$.

Theorem 2.1 (Index formula). *The multiplier $\rho_{p/q}(\varepsilon)$ of the q -cycle of $f_{\lambda_0 e^\varepsilon}$ bifurcating from 0 is holomorphic near $\varepsilon = 0$, and*

$$\rho_{p/q}(\varepsilon) = 1 - q^2 \varepsilon + q^3 \left(\iota_{p/q} - \frac{1}{2} \right) \varepsilon^2 + O(\varepsilon^3).$$

Equivalently, in (1), $r_0 = 1$, $r_1 = -1$ and $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$.

Proof. Choose a disc D around 0 on whose boundary $z \neq f^q(z)$ and which contains no other fixed point of f^q . By (2) and Rouché's theorem, for $|\varepsilon|$ small f_λ^q , $\lambda = \lambda_0 e^\varepsilon$, has exactly $q+1$ fixed points in D , counted with multiplicity. One of them is 0, with multiplier $\mu = \lambda^q = e^{q\varepsilon} \neq 1$, so it is simple. For each proper divisor d of q the map f^d has multiplier $\lambda_0^d \neq 1$ at 0, so f_λ^d has no fixed point in D other than 0. The remaining q points therefore have exact period q and form a single cycle; call its multiplier $\rho(\varepsilon)$. The cycle, as an unordered set, depends holomorphically on λ , hence so does ρ , a symmetric function of the cycle points. The function ρ is not identically 1, since the cycle is attracting on the satellite, so for $\varepsilon \neq 0$ small all $q+1$ fixed points are simple. Summing their indices $1/(1-m)$ gives

$$(3) \quad \frac{1}{1 - e^{q\varepsilon}} + \frac{q}{1 - \rho(\varepsilon)} = \frac{1}{2\pi i} \oint_{\partial D} \frac{dz}{z - f_\lambda^q(z)} \xrightarrow{\varepsilon \rightarrow 0} \iota_{p/q},$$

because the integrand converges uniformly on ∂D . The first term equals $-1/(q\varepsilon) + \frac{1}{2} + O(\varepsilon)$ and is unbounded. Hence $\rho(0) = 1$, and writing $1 - \rho = A\varepsilon^m(1 + O(\varepsilon))$ with $A \neq 0$, boundedness of the sum forces $m = 1$ and $A = q^2$. With $\rho = 1 - q^2\varepsilon + C\varepsilon^2 + O(\varepsilon^3)$ we get $q/(1 - \rho) = 1/(q\varepsilon) + C/q^3 + O(\varepsilon)$, and (3) yields $\frac{1}{2} + C/q^3 = \iota_{p/q}$. $\square$

Remark 2.2. The proof uses only that the parabolic point has multiplicity $q+1$. It therefore applies verbatim to the satellites of any hyperbolic component of $z^d + c$, whose parabolic points have a single cycle of petals. We checked this numerically for the period-two disc of M and for $z^3 + c$: the identity $r_2 = (\iota - \frac{1}{2})/q$ holds to 10^{-7} at $p/q = 2/7$ and $3/11$.

Remark 2.3 (Exact values). Since f^q has coefficients in $\mathbb{Z}[\lambda_0]$, the index is a rational function of λ_0 with rational coefficients, so $\iota_{p/q} \in \mathbb{Q}(\zeta_q)$ and $\iota_{p'/q} = \sigma(\iota_{p/q})$ for the automorphism $\sigma : \zeta_q \mapsto \zeta_q^{p'\bar{p}}$, $pp \equiv 1 \pmod{q}$. For example $\iota_{1/2} = \frac{1}{8}$, $\iota_{1/3} = \frac{100-8\sqrt{3}i}{441}$ and $\iota_{1/4} = \frac{1447-365i}{4624}$, so $\kappa(1/2) = -\frac{3}{16}$ and $\kappa(1/3) = \frac{-241-16\sqrt{3}i}{2646}$.

Definition 2.4. Let $c_{\text{root}} = c(\lambda_0)$, and let $c_{\text{ant}} \in \partial W_{p/q}$ be the unique boundary point at which the q -cycle has multiplier -1 . It is unique because the multiplier map $W_{p/q} \rightarrow \mathbb{D}$ is a conformal isomorphism in degree two. Set

$$d(p/q) = |c_{\text{ant}} - c_{\text{root}}|, \quad G(p/q) = \frac{d(p/q)}{2|c'(\lambda_0)|q^{-2}}.$$

Since $c(\lambda_0 e^{u/q^2}) - c(\lambda_0) = c'(\lambda_0)\lambda_0 u/q^2 + O(q^{-4})$, we have $G(p/q) = \frac{1}{2}|u_a| + O(q^{-2})$, where $R_{p/q}(u_a) = -1$. To first order $u_a = 2$ and $G = 1$; the deviation of G from 1 is governed at leading order by κ .

3. NORMAL FORMS AND THE ROTATION QUOTIENT

Theorem 3.1 (Normal form). *There is a unique formal conjugacy $H(z) = z + \sum_{k \geq 2} H_k z^k$ with $H_{q+1} = H_{2q+1} = 0$ such that $f \circ H = H \circ N$ up to order $2q + 1$, where*

$$N(z) = \lambda_0 z(1 + b_1 z^q + b_2 z^{2q}).$$

Put $\beta_i = \lambda_0 b_i$ and let $(H^2)_k$ denote the coefficient of z^k in $H(z)^2$. The coefficients are determined by the recursion

$$(4) \quad H_k(\lambda_0 - \lambda_0^k) = [k > q + 1] (k - q)\lambda_0^{k-q-1}\beta_1 H_{k-q} - (H^2)_k \quad (2 \leq k \leq 2q, \ k \neq q + 1),$$

together with $\beta_1 = (H^2)_{q+1}$ and $\beta_2 = (H^2)_{2q+1}$. Moreover

$$a(p/q) = q b_1, \quad q \iota_{p/q} = \frac{q^2 - 1}{2} + \frac{b_2}{b_1^2}, \quad \kappa(p/q) = \frac{1}{2} - \frac{1}{2q} - \frac{1}{2q^2} + \frac{b_2}{(qb_1)^2}.$$

Proof. Comparing coefficients of z^k in $\lambda_0 H + H^2 = \sum_m H_m N^m$ gives (4). We use that $[N^m]_k$ is nonzero only for $k \in \{m, m+q, m+2q\}$, with values λ_0^m and $m\lambda_0^{m-1}\beta_1$ in the first two cases. The factor $\lambda_0 - \lambda_0^k$ vanishes exactly when $k \equiv 1 \pmod{q}$. At $k = q + 1$ and $k = 2q + 1$ the equation fixes β_1, β_2 and leaves H_{q+1}, H_{2q+1} free; we set them to zero. For the second part, write $N^j(z) = \lambda_0^j z(1 + A_j z^q + B_j z^{2q} + \dots)$. Using $\lambda_0^q = 1$ and $(N^j)^q = z^q(1 + qA_j z^q + \dots)$,

$$A_{j+1} = A_j + b_1, \quad B_{j+1} = B_j + (q + 1)b_1 A_j + b_2,$$

so $A_q = qb_1$ and $B_q = \frac{(q+1)q(q-1)}{2}b_1^2 + qb_2$. The coefficient a and the index are invariant under formal conjugacy tangent to the identity; for $z + Az^{q+1} + Bz^{2q+1}$ the index is B/A^2 . This gives the formulas for a and ι , and the formula for κ follows from Theorem 2.1. $\square$

Corollary 3.2 (κ is an index). *Since N commutes with $z \mapsto \lambda_0 z$, the rotation quotient $\Phi_{p/q}(W) := N^q(W^{1/q})^q = W + q^2 b_1 W^2 + \dots$ is a well-defined germ, and*

$$\kappa(p/q) = \iota(\Phi_{p/q}) - \frac{1}{2} = \frac{1}{2} - \text{résit}(\Phi_{p/q}).$$

Proof. From $N^q(z) = z(1 + A_q z^q + B_q z^{2q} + \dots)$ we get $\Phi(W) = W(1 + A_q W + B_q W^2)^q + \dots$. Hence $\Phi = W + AW^2 + BW^3 + \dots$ with $A = qA_q = q^2 b_1$ and $B = qB_q + \binom{q}{2}A_q^2$. Then $\iota(\Phi) = B/A^2 = 1 - \frac{1}{2q} - \frac{1}{2q^2} + \frac{b_2}{(qb_1)^2}$, which is $\kappa + \frac{1}{2}$ by Theorem 3.1. $\square$

Remark 3.3 (Computation). The recursion (4) costs $O(q^2)$ arithmetic operations, and it does not suffer cancellation. At $q = 128$ and $q = 301$ the computed κ is unchanged in more than twenty digits when the working precision is raised from 60 to over 300 digits. Composing f symbolically q times instead costs $O(q^3)$ and loses about q decimal digits to cancellation at $p = 1$, where $|a(1/q)| \approx e^{2.3q}$. We used the latter, in ball arithmetic [9], only as a certified cross-check (Appendix A).

q	$\kappa(1/q)$	$ \kappa(1/q) + \frac{i}{2\pi q} - \kappa_0 $
64	$0.02385454027565757 - 0.05408067667475267 i$	$7.1 \cdot 10^{-4}$
256	$0.02383094323212187 - 0.05288194473715776 i$	$4.5 \cdot 10^{-5}$
1024	$0.02382625451962614 - 0.05245731336604759 i$	$2.8 \cdot 10^{-6}$
4096	$0.02382591114406885 - 0.05234334224894280 i$	$1.7 \cdot 10^{-7}$
8192	$0.02382589337085374 - 0.05232404394907701 i$	$4.4 \cdot 10^{-8}$
16384	$0.02382588889851146 - 0.05231436234675194 i$	$1.1 \cdot 10^{-8}$
32768	$0.02382588777679670 - 0.05230951343396723 i$	$2.7 \cdot 10^{-9}$
fit, $q \rightarrow \infty$	$0.02382588740220057 - 0.05230465911400367 i$	
horn map, Thm. 5.2	$0.02382588740220057 - 0.05230465911400367 i$	

TABLE 1. $\kappa(1/q)$ from the normal-form recursion (first sixteen digits shown). The corrected error decays like q^{-2} . Values for $q \geq 1024$ are certified (Arb, radius $\leq 5.4 \cdot 10^{-44}$). The fit uses the thirteen values $256 \leq q \leq 32768$ with eleven terms in $1/q$ (residual $3.6 \cdot 10^{-29}$); the last row is the horn-map value of Section 5.

4. THE LIMIT κ_0

Numerical result 4.1. Table 1 lists $\kappa(1/q)$ for q up to 32768; for $q \geq 1024$ the values are certified in ball arithmetic (Appendix A). Figure 1 shows the convergence. Least-squares fits of $\kappa(1/q) = \sum_{j \leq J} c_j q^{-j}$ over $q \geq q_{\min}$ are stable under changes of $(q_{\min}, J)$. The best overdetermined fit, $(q_{\min}, J) = (256, 10)$ on thirteen values with residual $3.6 \cdot 10^{-29}$, gives

$$c_0 = 0.023825887402200569000729189645 \dots - 0.052304659114003666316522486208 \dots i$$

(28 digits shown), and the first-order coefficient

$$c_1 = -\frac{i}{2\pi} \quad \text{to within } 2 \cdot 10^{-23},$$

and $c_2 = 0.402776138694 + 2.902726170410 i$. The ratios $|c_{j+1}/c_j|$ stay near 7, consistent with an expansion convergent for $q \gtrsim 7$.

5. THE HORN MAP OF $z + z^2$

We recall the objects of parabolic implosion [11, 16, 8, 10] in the concrete form we compute with. In the coordinate $w = -1/z$ the map $f_0(z) = z + z^2$ becomes

$$F(w) = \frac{w^2}{w-1} = w + 1 + \frac{1}{w-1}.$$

Lemma 5.1 (Formal Fatou coordinate). *There is a unique formal series $\Phi(w) = w - \log w + \sum_{k \geq 1} c_k w^{-k}$ with $\Phi \circ F = \Phi + 1$. Its coefficients are rational:*

$$c_1, c_2, \dots = \frac{1}{2}, \frac{1}{3}, \frac{13}{36}, \frac{113}{240}, \frac{1187}{1800}, \frac{877}{945}, \dots$$

With $u = 1/w$ they satisfy the triangular system

$$\sum_{k \geq 1} c_k [(u - u^2)^k - u^k] = - \sum_{m \geq 2} \left(1 - \frac{1}{m}\right) u^m.$$

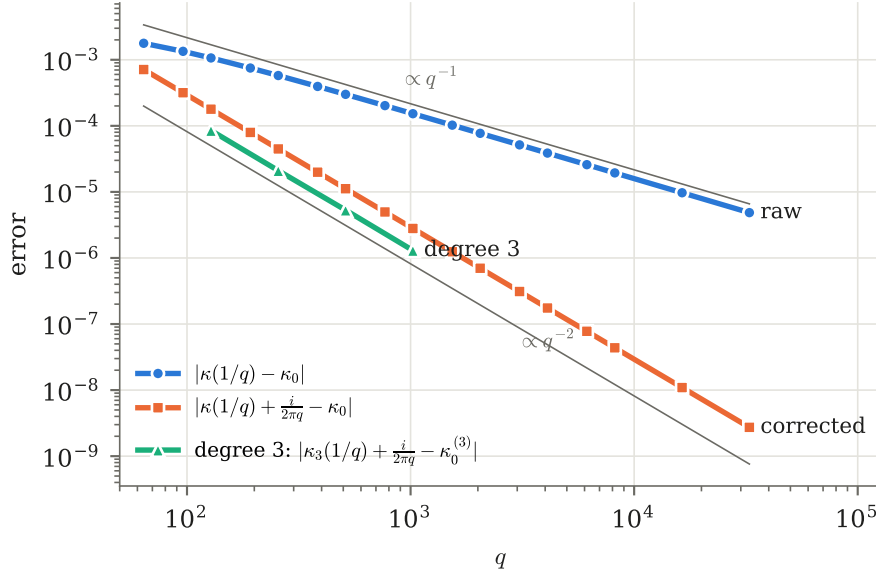


FIGURE 1. Convergence of $\kappa(1/q)$ to κ_0 (degree two) and of $\kappa_3(1/q)$ to $\kappa_0^{(3)}$ (degree three, Section 7). The raw error decays like q^{-1} ; after the universal correction $-i/(2\pi q)$ it decays like q^{-2} . Grey lines are slope guides.

Proof. With $u = 1/w$ we have $1/F = u - u^2$, $F - w - 1 = u/(1 - u)$ and $\log F - \log w = -\log(1 - u)$. Substituting gives the displayed identity. The coefficient of u^m contains c_{m-1} with factor $-(m-1)$ and otherwise only c_k with $k < m-1$. The coefficient of $\log w$ is fixed by the order- u term, and equals the résidu itératif 1 of f_0 . $\square$

Let $\Phi_{\text{att}}(w) = \lim_n (\Phi(F^n(w)) - n)$ on the attracting petal, using the principal logarithm. Let $\Psi_{\text{rep}}(Z) = \lim_n F^n(\Phi^{-1}(Z - n))$ be the repelling Fatou parametrization, using the branch $\arg w \in (0, 2\pi)$ on the left. The upper horn map $E = \Phi_{\text{att}} \circ \Psi_{\text{rep}}$ commutes with $Z \mapsto Z + 1$ and, for $\text{Im } Z$ large, has an expansion

$$E(Z) = Z + a_0 + \sum_{n \geq 1} a_n e^{2\pi i n Z}.$$

Replacing Φ_{att} or Ψ_{rep} by a translate changes a_0 , and conjugates E by $Z \mapsto Z + c$, which sends $a_n \mapsto a_n e^{2\pi i n c}$. In $W = e^{2\pi i Z}$ the horn map is the germ $W \mapsto e^{2\pi i a_0} \mathcal{P}_0(W)$, where

$$\mathcal{P}_0(W) := W \exp\left(2\pi i \sum_{n \geq 1} a_n W^n\right)$$

is the parabolic renormalization of f_0 normalized to multiplier one [10, 8].

Theorem 5.2. $\iota(\mathcal{P}_0) = \frac{1}{2} + \frac{a_2}{2\pi i a_1^2}$. This quantity does not depend on the normalizations of the Fatou coordinates.

Proof. $\mathcal{P}_0(W) = W + AW^2 + BW^3 + \dots$ with $A = 2\pi i a_1$ and $B = 2\pi i a_2 + \frac{1}{2}(2\pi i a_1)^2$, and $\iota = B/A^2$. Under $a_n \mapsto a_n e^{2\pi i n c}$ the ratio a_2/a_1^2 is unchanged; a_0 does not enter. $\square$

Numerical result 5.3. Computing E on the line $\text{Im } Z = \frac{1}{4}$ at 60 and at 80 digits (Appendix A) gives $a_0 = 0$ to working precision ($|a_0| \approx 10^{-58}$ at 60 digits) and

$$\begin{aligned} \text{Re } a_1 &= -0.05895043056440687000807026251551, \\ \text{Im } a_1 &= 0.010009865898273724511535250164253, \\ \text{Re } a_2 &= 0.0012858196663089284889517464706577, \\ \text{Im } a_2 &= 0.00011738770718282661923962731179377, \end{aligned}$$

with $|a_n|$ decreasing by a factor of about 55 per step. Hence

$$\begin{aligned} \text{Re}(\iota(\mathcal{P}_0) - \tfrac{1}{2}) &= 0.023825887402200569000729189645402, \\ \text{Im}(\iota(\mathcal{P}_0) - \tfrac{1}{2}) &= -0.052304659114003666316522486207571. \end{aligned}$$

The two precision settings agree in all 32 digits shown. This coincides with c_0 of Numerical result 4.1 to $3.1 \cdot 10^{-28}$; other stable fits differ from it by at most $5 \cdot 10^{-25}$. We therefore claim agreement to at least 24 digits, and 27 on the best fit. Already in double precision the agreement is $2.5 \cdot 10^{-11}$ at every height $\text{Im } Z \in [\frac{1}{4}, \frac{1}{2}]$.

Conjecture 5.4. The limit $\kappa_0 = \lim_{q \rightarrow \infty} \kappa(1/q)$ exists and equals $\iota(\mathcal{P}_0) - \frac{1}{2} = \frac{a_2}{2\pi i a_1^2}$. Moreover

$$\kappa(1/q) = \kappa_0 - \frac{i}{2\pi q} + O(q^{-2}).$$

6. RENORMALIZATION INTERPRETATION AND FURTHER EVIDENCE

6.1. Heuristic. For α near 0, the near-parabolic renormalization $\mathcal{R}$ of Inou and Shishikura [8] assigns to $f_\alpha(z) = e^{2\pi i \alpha} z + z^2$ a germ $\mathcal{R}f_\alpha$ at 0 with multiplier $e^{-2\pi i/\alpha}$. By parabolic implosion [11, 16], as $\alpha \rightarrow 0$ with $-1/\alpha \rightarrow \sigma \pmod{\mathbb{Z}}$, the germ $\mathcal{R}f_\alpha$ tends to $e^{2\pi i \sigma}$ times the horn germ. For $\alpha = 1/q$ the phase is 0, and $\mathcal{R}f_{1/q}$ is a parabolic germ with multiplier one. The rotation quotient $\Phi_{1/q}$ of Corollary 3.2 is also a parabolic germ with multiplier one, obtained from f^q by passing to the quotient by the q -fold rotation. If $\Phi_{1/q}$ and $\mathcal{R}f_{1/q}$ are conformally conjugate, Corollary 3.2 and continuity of $\mathcal{R}$ give Conjecture 5.4. The same reasoning leads to two further predictions, which we test next.

6.2. Bounded p . Fix $p \geq 2$ and r with $\gcd(p, r) = 1$, and let $q = pN + r \rightarrow \infty$. Then $-1/\alpha = -q/p \equiv -r/p \pmod{1}$, so $\mathcal{R}f_{p/q} \rightarrow \mu \mathcal{P}_0$ with $\mu = e^{-2\pi i r/p}$. This germ has multiplier a primitive p -th root of unity. Applying Corollary 3.2 to it predicts

$$(5) \quad \lim_{N \rightarrow \infty} \kappa\left(\frac{p}{pN + r}\right) = \frac{\iota((\mu \mathcal{P}_0)^p) - \frac{1}{2}}{p}, \quad \mu = e^{-2\pi i r/p}.$$

Table 2 compares (5) with bulb data for all nine classes with $p \leq 5$. The bulb values are normal-form values of $\kappa(p/q)$ for $N \leq 1024$, extrapolated in $1/q$. All nine agree to between $6 \cdot 10^{-9}$ and $2.6 \cdot 10^{-8}$, within the extrapolation uncertainty of about 10^{-7} . The opposite phase convention, $\mu = e^{+2\pi i r/p}$, exchanges the classes r and $p - r$ and fails. The first-order coefficients satisfy $c_1(p) \cdot 2\pi p / (-i) = 1.0008, 1.0011, 1.0014, 1.0016$ for $p = 2, \dots, 5$, suggesting $c_1(p) = -i/(2\pi p)$ in general.

p	r	horn-map prediction $(\iota((\mu\mathcal{P}_0)^p) - \frac{1}{2})/p$	bulbs, extrapolated	$ \Delta $
2	1	$0.049693182903 - 0.023894096844i$	$0.049693177608 - 0.023894071809i$	$2.6 \cdot 10^{-8}$
3	1	$0.054529447219 - 0.0050171894089i$	$0.054529443083 - 0.0050171743067i$	$1.6 \cdot 10^{-8}$
3	2	$0.055036036944 - 0.023467983453i$	$0.055036033035 - 0.023467969651i$	$1.4 \cdot 10^{-8}$
4	1	$0.053737282786 + 0.0074895906470i$	$0.053737279652 + 0.0074896009583i$	$1.1 \cdot 10^{-8}$
4	3	$0.054739672309 - 0.026348641547i$	$0.054739669177 - 0.026348633046i$	$9.1 \cdot 10^{-9}$
5	1	$0.051130771685 + 0.015984969447i$	$0.051130769387 + 0.015984976959i$	$7.9 \cdot 10^{-9}$
5	2	$0.059925460063 - 0.010450443803i$	$0.059925457760 - 0.010450436333i$	$7.8 \cdot 10^{-9}$
5	3	$0.060335706660 - 0.0045365352268i$	$0.060335704038 - 0.0045365278901i$	$7.8 \cdot 10^{-9}$
5	4	$0.052624850350 - 0.029174853238i$	$0.052624847638 - 0.029174847754i$	$6.1 \cdot 10^{-9}$

TABLE 2. Bounded- p limits of κ (degree two): the horn-map prediction (5) against the bulb values extrapolated from $q = pN + r$, $64 \leq N \leq 1024$.

p	r	G at $q = 512p + r$	G at $q = 2048p + r$	extrapolated	horn map
1	1	1.02634395	1.02639633	1.02641535	1.02641539
2	1	1.11142282	1.11143038	1.11143322	1.11143318
3	1	1.12836657	1.12836757	1.12836781	1.12836776
3	2	1.12537076	1.12537450	1.12537590	1.12537582
4	1	1.12357397	1.12357381	1.12357368	1.12357348
4	3	1.12156857	1.12157111	1.12157204	1.12157195
5	1	1.11300416	1.11300362	1.11300313	1.11300337
5	2	1.14414695	1.14414738	1.14414703	1.14414795
5	3	1.14644523	1.14644563	1.14644539	1.14644595
5	4	1.11312789	1.11312962	1.11312984	1.11313044

TABLE 3. Bulb size G (Definition 2.4) along $q = pN + r$, extrapolated from $N = 512, 1024, 2048$, against the prediction from $M_u = \mu e^{u/p^2} \mathcal{P}_0$. The first row gives $G_0 = \lim G(1/q)$.

6.3. The unfolding and the bulb size. With $\lambda = \lambda_0 e^{u/q^2}$ one has $\alpha = \frac{p}{q} + \frac{u}{2\pi i q^2}$, so $e^{-2\pi i/\alpha} = \mu e^{u/p^2} + O(q^{-1})$. The picture therefore predicts that $R_{p/q}(u)$ converges to the multiplier $\rho_{p,r}(u)$ of the p -cycle of

$$M_u(W) = \mu e^{u/p^2} \mathcal{P}_0(W)$$

that bifurcates from $W = 0$.

For $p = 1$ the Taylor coefficients of $\rho_{1,1}$ at $u = 0$ agree with those of $R_{1/1025}$ through order eight, up to the finite- q shift. For instance $r_3 = -3.78 \cdot 10^{-4} - 9.80 \cdot 10^{-4}i$ against $-3.62 \cdot 10^{-4} - 9.72 \cdot 10^{-4}i$, and $r_8 = 3.14 \cdot 10^{-9} - 4.56 \cdot 10^{-9}i$ against $3.13 \cdot 10^{-9} - 4.50 \cdot 10^{-9}i$. Here r_3 depends on the unfolding and not only on the germ, so this checks that u enters as the Lavaurs phase. The finite- q shifts themselves are explained in Section 6.4. Solving $\rho_{p,r}(u_a) = -1$ gives a prediction for $\lim G$. Table 3 compares it with the bulb sizes for all ten classes with $p \leq 5$; the agreement is at most $9 \cdot 10^{-7}$.

6.4. The first-order correction is kinematic. The unfolding above uses only the leading term $e^{-2\pi i/\alpha} \approx \mu e^{u/p^2}$. Exactly, the exponent is a Möbius function of u .

Proposition 6.1. *Let $\lambda = e^{2\pi i \alpha} = \lambda_0 e^{u/q^2}$ with $\lambda_0 = e^{2\pi i p/q}$. Then $e^{-2\pi i/\alpha} = \mu e^{\tilde{u}/p^2}$ with $\mu = e^{-2\pi i q/p}$ and*

$$\tilde{u} = \frac{u}{1 + u/(2\pi i p q)} = u + \frac{i u^2}{2\pi p q} + O\left(\frac{u^3}{q^2}\right).$$

Proof. $\alpha = (2\pi i p q + u)/(2\pi i q^2)$, so $-1/\alpha + q/p = q u/(p(2\pi i p q + u))$, and $2\pi i p^2$ times this is $\tilde{u}$. $\square$

If the renormalized germ depends on α only through this multiplier, up to $O(q^{-2})$, then $R_{p/q}(u) = \rho_{p,r}(\tilde{u}(u)) + O(q^{-2})$ coefficientwise. Comparing the u^2 and u^3 coefficients, with $\rho_{p,r}(u) = 1 - u + \kappa_{p,r}u^2 + \rho_3u^3 + \dots$, gives

$$(6) \quad \kappa(p/q) = \kappa_{p,r} - \frac{i}{2\pi p q} + O(q^{-2}), \quad r_3(p/q) = \rho_3 + \frac{i \kappa_{p,r}}{\pi p q} + O(q^{-2}).$$

The first identity is the observed coefficient $c_1(p) = -i/(2\pi p)$ of Conjecture 5.4 and Section 6.2. It involves no property of the germ, which explains why it is the same in degree three.

Numerical result 6.2. Let $r_k(p/q)$ be the Taylor coefficients of the bulb (Cauchy integrals on $|u| = 1.5$, double precision) and ρ_k those of the renormalized family M_u (Cauchy integrals on the same circle). For $2 \leq k \leq 8$, $(p, r) \in \{(1, 1), (2, 1), (3, 1), (3, 2)\}$ and $q = pN + r$ with $N = 64, 128, \dots, 1024$, the residuals $|r_k(p/q) - [u^k]\rho_{p,r}(\tilde{u})|$ decrease by a factor between 3.77 and 4.28 per doubling of N , in all 41 cases where the residual exceeds 10^{-8} ; below that level the round-off of the Cauchy integrals takes over. Without the substitution $u \mapsto \tilde{u}$ the factor lies between 1.1 and 2.0. In degree three, with the cubic horn germ, Cauchy integrals on $|u| = 1$ and $p = 1$, the factor lies between 3.75 and 4.06 (26 cases).

So through first order in $1/q$ the whole function $R_{p/q}$, not only κ , is the renormalized family composed with the exact Inou-Shishikura phase: the germ itself has no $O(1/q)$ correction. For $p = 1$, $q = 1025$, (6) predicts $r_3 - \rho_3 = 1.62 \cdot 10^{-5} + 0.74 \cdot 10^{-5}i$, while the values in Section 6.3 give $1.6 \cdot 10^{-5} + 0.8 \cdot 10^{-5}i$.

6.5. The hierarchy. For $\alpha = [0; a_1, a_2, \dots]$ the renormalization $\mathcal{R}f_\alpha$ has multiplier $e^{-2\pi i/\alpha}$. Conjugating by $z \mapsto \bar{z}$ turns its rotation number into $1/\alpha \bmod 1 = [0; a_2, a_3, \dots]$, so that $\mathcal{R}$ acts on continued fractions as the Gauss map [8]. When the first k partial quotients tend to infinity, the picture therefore predicts that each renormalization is close to $\mathcal{P}_0$ applied to the complex conjugate of the previous one. Write $\bar{G}(W) = \overline{G(\bar{W})}$, normalize germs to the form $v + v^2 + O(v^3)$, and set

$$G_1 = \mathcal{P}_0, \quad G_{k+1} = \mathcal{P}(\bar{G}_k), \quad \kappa_k = \iota(G_k) - \frac{1}{2},$$

where $\mathcal{P}$ denotes the upper-horn-map construction of Section 5 applied to an arbitrary parabolic germ, so that $\kappa_k = a_2/(2\pi i a_1^2)$ for the horn map of $\bar{G}_{k-1}$.

The case $k = 2$ can be tested against bulbs. By (5) with $r = 1$, the limit of $\kappa([0; N, p])$ as $N \rightarrow \infty$ is an index of $(e^{-2\pi i/p} \mathcal{P}_0)^p$. We compute this index for $p \leq 256$ with the normal form of Theorem 3.1 generalized to arbitrary germs (an $O(p^3)$ recursion) and extrapolate in $1/p$. The result, $0.0112359330 + 0.0377814233i$, equals $\bar{\kappa}_2$ to $1.2 \cdot 10^{-8}$. The conjugate appears because $p^{-1} \bmod q \rightarrow 0^-$ along this sequence, against 0^+ for $p = 1$. Without the conjugation in the definition of G_2 one obtains $0.014770 - 0.033911i$, which is excluded by the data.

For $k \geq 3$ the germs G_k are given by convergent series with small radius (about 2.6 in the normalized coordinate v at $k = 2$), and the horn maps must be sampled at heights where the orbits stay well inside the disc. Table 4 lists $\kappa_1, \dots, \kappa_7$ for two independent settings of height, Fourier truncation and precision. They agree to $2 \cdot 10^{-14}$ or better at every level.

k	κ_k ($h = 1$)	$ \kappa_k - \kappa_{k-1} $	$ \kappa_k^{(h=1)} - \kappa_k^{(h=5/4)} $
1	0.0238258874022 - 0.0523046591140 i		—
2	0.0112359215880 - 0.0377814219023 i	$1.9 \cdot 10^{-2}$	$6.1 \cdot 10^{-16}$
3	0.0119510972606 - 0.0374792854625 i	$7.8 \cdot 10^{-4}$	$1.7 \cdot 10^{-14}$
4	0.0119279615825 - 0.0374560782080 i	$3.3 \cdot 10^{-5}$	$5.8 \cdot 10^{-16}$
5	0.0119292218902 - 0.0374555354669 i	$1.4 \cdot 10^{-6}$	$2.5 \cdot 10^{-17}$
6	0.0119291812682 - 0.0374554947782 i	$5.7 \cdot 10^{-8}$	$9.9 \cdot 10^{-19}$
7	0.0119291834806 - 0.0374554938253 i	$2.4 \cdot 10^{-9}$	$1.2 \cdot 10^{-19}$
∞	0.011929183413 - 0.037455493752 i		

TABLE 4. The hierarchy $\kappa_k = \iota(G_k) - \frac{1}{2}$, $G_{k+1} = \mathcal{P}(\bar{G}_k)$, computed at heights $h = 1$ (16 Fourier modes, 90 digits) and $h = \frac{5}{4}$ (14 modes, 100 digits). Level 1 is κ_0 and is common to both. The last row is the real-linear extrapolation described in Numerical result 6.3.

Numerical result 6.3. The differences $\Delta_k = \kappa_k - \kappa_{k-1}$ decrease geometrically, $|\Delta_{k+1}/\Delta_k| \rightarrow 0.0419$, with ratios $\Delta_{k+1}/\Delta_k \approx -0.01546 \mp 0.03894i$ alternating in sign of the imaginary part. This is the signature of an antilinear contraction $\Delta \mapsto \nu\bar{\Delta}$, the linearization of $G \mapsto \mathcal{P}(\bar{G})$ being antilinear. Fitting $\Delta_{k+1} = M\Delta_k$ with M real-linear on $\mathbb{C} \cong \mathbb{R}^2$ gives eigenvalues ± 0.041899 and the limit

$$\kappa_* := \lim_{k \rightarrow \infty} \kappa_k = 0.011929183413 - 0.037455493752i,$$

with the fits through levels 6 and 7 differing by less than $2 \cdot 10^{-12}$. Thus $\kappa_* + \frac{1}{2}$ is the cubic coefficient of the normalized fixed point $G_* = \mathcal{P}(\bar{G}_*)$.

Remark 6.4. Without the conjugation, $G \mapsto \mathcal{P}(G)$ is the parabolic renormalization operator of Inou–Shishikura, whose fixed point f_* was computed by Lanford and Yampolsky [10]: $f_*(z) \approx z + z^2 + (0.514 - 0.0346i)z^3$, with leading eigenvalue $\approx -0.017 + 0.040i$ of $D\mathcal{P}$ at f_* . Running our code without conjugation reproduces both: the cubic coefficients converge to $0.5142768 - 0.0346276i$ (extrapolated, $\pm 10^{-7}$), with ratio of successive differences $\rightarrow -0.01764 + 0.04038i$. The bulbs, however, select the fixed point $G_* \neq f_*$ of the conjugated operator, whose cubic coefficient is $\kappa_* + \frac{1}{2} = 0.5119292 - 0.0374555i$.

Conjecture 6.5. For every $k \geq 1$,

$$\lim_{N_k \rightarrow \infty} \cdots \lim_{N_1 \rightarrow \infty} \kappa([0; N_1, \dots, N_k]) = \begin{cases} \kappa_k, & k \text{ odd,} \\ \bar{\kappa}_k, & k \text{ even.} \end{cases}$$

In particular κ_* and $\bar{\kappa}_*$ are the limits of the first correction as the number of large partial quotients of p/q grows, taken along odd and even k respectively.

The case $k = 1$ is Numerical result 5.3, and $k = 2$ is the iterated limit discussed above. We also tested the diagonal sequences $[0; N, \dots, N]$, computing κ with the normal form in midpoint arithmetic at 128 and 192 bits (the two agree to $4 \cdot 10^{-35}$) and fitting polynomials in $1/N$. For $k = 2$ and $16 \leq N \leq 160$ ($q \leq 25601$) the limit agrees with $\bar{\kappa}_2$ to $3 \cdot 10^{-10}$, so the diagonal and the iterated limit coincide. For $k = 3$ and $6 \leq N \leq 40$ ($q \leq 64080$) the limit is $0.0119509937 - 0.0374804294i$. This agrees with κ_3 to $1.2 \cdot 10^{-6}$, and every fit with $N \geq 10$ and 5 or 6 terms lies within $1.4 \cdot 10^{-5}$ of κ_3 , whereas $|\kappa_3 - \kappa_2| = 7.8 \cdot 10^{-4}$ and $|\kappa_3 - \bar{\kappa}_3| = 7.5 \cdot 10^{-2}$. Thus the conjecture, including its parity rule, holds numerically for $k \leq 3$. Level 4 ($|\kappa_4 - \kappa_3| = 3.3 \cdot 10^{-5}$) would require periods $q \approx N^4$ with $N \gtrsim 30$.

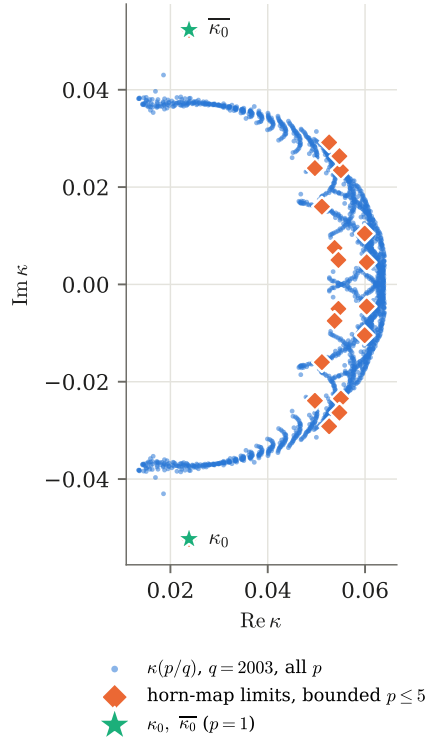


FIGURE 2. All $\kappa(p/q)$ for $q = 2003$ (values for $p > q/2$ from the symmetry $\kappa(1 - p/q) = \overline{\kappa(p/q)}$), the horn-map limits (5) for $p \leq 5$, and $\kappa_0, \overline{\kappa_0}$. The $p = 1$ bulbs are the extreme points of the arithmetic cloud.

7. DEGREE THREE: A SECOND UNIVERSALITY CLASS

For $z^3 + c$ the main component is parametrized by the multiplier λ of the fixed point $z = (\lambda/3)^{1/2}$, with $c = z - z^3$. Its cusp at $\lambda = 1$, where $z_* = 1/\sqrt{3}$ and $c_* = 2/(3\sqrt{3})$, has the parabolic germ $w \mapsto w + \sqrt{3}w^2 + w^3$. In $v = \sqrt{3}w$ this is

$$g(v) = v + v^2 + \frac{1}{3}v^3,$$

whose résidu itératif is $1 - \frac{1}{3} = \frac{2}{3}$. Lemma 5.1 generalizes with $F(w) = -1/g(-1/w)$ and $\Phi(w) = w - \frac{2}{3} \log w + \sum c_k w^{-k}$, where $c_1, c_2, \dots = \frac{2}{9}, \frac{5}{54}, \frac{14}{243}, \frac{89}{2430}, \dots$. The horn map gives, at 50 and at 70 digits,

$$\kappa_0^{(3)} := \frac{a_2}{2\pi i a_1^2}, \quad \begin{aligned} \operatorname{Re} \kappa_0^{(3)} &= 0.143599278095291569488064104088, \\ \operatorname{Im} \kappa_0^{(3)} &= -0.030311647898502537611943110858. \end{aligned}$$

Table 5 shows that $\kappa_3(1/q) + \frac{i}{2\pi q}$ converges to this value like q^{-2} . So the correction $-i/(2\pi q)$ is the same in degree three, while the constant is not. The bounded- p predictions hold in degree three as well: κ agrees to within $5 \cdot 10^{-6}$ for $(p, r) = (1, 1), (2, 1), (3, 1), (3, 2)$ at $q \approx 512p$.

One subtlety concerns the bulb size. At the centre of a degree- d satellite the critical point lies on the cycle, so the multiplier $\prod d z_j^{d-1}$ vanishes to order $d-1$. The multiplier map $W \rightarrow \mathbb{D}$ is proper, since the cycle becomes indifferent at ∂W , and each bounded hyperbolic component is a cell with a unique centre [13]. So the multiplier map is a branched cover of degree $d-1$, and

q	$\kappa_3(1/q)$	$ \kappa_3(1/q) + \frac{i}{2\pi q} - \kappa_0^{(3)} $
128	$0.1436180525 - 0.03147252376i$	$8.5 \cdot 10^{-5}$
256	$0.1436043106 - 0.03091278633i$	$2.1 \cdot 10^{-5}$
512	$0.1436005784 - 0.03061736685i$	$5.3 \cdot 10^{-6}$
1024	$0.1435996089 - 0.03046578967i$	$1.3 \cdot 10^{-6}$
horn map of $v + v^2 + \frac{1}{3}v^3$		$0.1435992781 - 0.03031164790i$

TABLE 5. Degree three: $\kappa_3(1/q)$ from Cauchy integrals on $|u| = 1$ (Appendix A) against the horn map of the cubic cusp germ.

in degree three there are two points with multiplier -1 . Both are predicted: at $p = 3$, $r = 1$ the horn map gives $G = 1.196074$ and 1.227092 , against extrapolated bulb values 1.196075 and 1.227091 .

8. ARITHMETIC NATURE

We searched with PSLQ [4], at 30 digits and coefficient bound 1000, for integer relations between each of $\operatorname{Re} \kappa_0$, $\operatorname{Im} \kappa_0$, $|\kappa_0|^2$, $\arg \kappa_0/\pi$, $\operatorname{Re}(2\pi i \kappa_0)$, $\operatorname{Im}(2\pi i \kappa_0)$, $|a_1|$ and $\arg a_1/\pi$, and every subset of at most three of $\{1, \pi, \pi^2, 1/\pi, \log 2, \zeta(3), G, \gamma\}$, where G is Catalan's constant. No relation was found. This is consistent with the general expectation that invariants of parabolic germs, such as Écalle–Voronin moduli [3, 17], are transcendental numbers without closed form. We regard κ_0 as a new constant of this kind, attached to the quadratic family and to the Mandelbrot set.

9. DISCUSSION

The first-order theory of satellite bulbs is the Ford-circle picture, with radii q^{-2} and Farey tangencies. The first correction, $\kappa(p/q)$, is the holomorphic index of a parabolic germ (Theorem 2.1) and equals the index of the rotation quotient minus $\frac{1}{2}$ (Corollary 3.2). Numerically, it is controlled by near-parabolic renormalization. The constant κ_0 is the value at the fixed point of the simplest instance, $\alpha = 1/q \rightarrow 0$. Bounded p corresponds to one renormalization step with phase $e^{-2\pi i r/p}$. General p/q should correspond to iterating renormalization along the continued fraction of p/q ; Section 6.5 carries this out when all partial quotients are large, where the iteration contracts by a factor 0.042 per step. Hyperbolicity of $\mathcal{R}$ would then explain why $\kappa(p/q)$ depends mainly on the last partial quotients, that is on $p^{-1} \bmod q$ [15].

Open problems.

- (1) Prove Conjecture 5.4, for instance by establishing a conformal conjugacy between $\Phi_{1/q}$ and $\mathcal{R}f_{1/q}$ together with continuity of $\mathcal{R}$ at integer $1/\alpha$.
- (2) Prove that the renormalized germ has no $O(1/q)$ correction beyond the phase of Proposition 6.1 (Numerical result 6.2); by (6) this gives the universal term $-i/(2\pi p q)$.
- (3) Prove Conjecture 6.5 (verified numerically for $k \leq 3$), and identify the fixed point G_* of $G \mapsto \mathcal{P}(\bar{G})$ and its spectrum.
- (4) Compute the generic limit function $\kappa(p/q)$, $p \rightarrow \infty$, by iterated renormalization with bounded partial quotients, including its values at quadratic irrationals as periodic points of $\mathcal{R}$.
- (5) Decide whether κ_0 can be expressed through Écalle–Voronin invariants of $z + z^2$ in closed form.

APPENDIX A. NUMERICAL METHODS AND ERROR CONTROL

A.1 Normal form. Recursion (4) was run in `mpmath` [14] at 50 working digits for $64 \leq q \leq 8192$ ($p = 1$) and $q \leq 5124$ (bounded p). It was also implemented in Arb ball arithmetic [9], which certifies every value. There the online convolution $(H^2)_k$ is evaluated in blocks: completed blocks are folded in by truncated series products, and the remainder is summed directly. The working precision is doubled until the certified radius of κ is below 10^{-40} . Ball radii are pessimistic, since they cannot see cancellation, but they remain small: for $p = 1$ they are $1.6 \cdot 10^{-50}$ at $q = 1024$ and below $2 \cdot 10^{-111}$ at $q = 32768$. The two implementations agree to the 40 digits stored. The ball version is 60–110 times faster; $q = 32768$ takes about two minutes.

A.2 Extrapolation. Limits in q come from least-squares fits in powers of $1/q$ at 40–60 digits. Each is reported only when it is stable, to the stated precision, under changes of the fit order and of the smallest q used.

A.3 Horn maps. The Fatou series were truncated at $K = 40$ terms and evaluated at $|w| \geq 800$. At that radius the truncation error is below 10^{-60} , since $|c_k|$ grows roughly like $k!/(2\pi)^k$. We used $n = 1000$ – 1800 repelling iterations and computed the Fourier coefficients from 64–80 samples on $\text{Im } Z = \frac{1}{4}$, with aliasing below 10^{-58} . Two independent settings agree to all digits reported. The coefficients a_n extracted at height h carry noise amplified by $e^{2\pi nh}$, so for the unfolding computations we used $n \leq 20$ coefficients computed at 60 digits.

A.4 Multiplier data. The coefficients r_k were also computed by Cauchy integrals of $R_{p/q}$ over a circle, continuing the q -cycle numerically. For $z^2 + c$ the radius $|u| = 2.5$ is admissible. For $z^3 + c$ the function $R_{p/q}$ has a singularity inside $|u| < 2.5$, and the radius must be at most 1.5. The condition $r_1 = -1$ from Theorem 2.1 is a reliable diagnostic: it fails as $r_1 = -1.032$ on the larger circle.

A.5 Reproducibility. All code, data and tests are in the accompanying repository [15]. The package `bulbford` contains `normal_form` and `normal_form_ball`, `horn`, `germ`, `renorm` and `dynamics`; the cycle iteration in `dynamics` is compiled with Numba. The tables and figures of this paper are generated by `paper/tables.py` and `paper/figures.py`. The test suite checks Theorems 2.1 and 3.1, the horn-map value of κ_0 to 25 digits, and the bounded- p and degree-three predictions.

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